

EMERALD, Australia

WMO: 943630

Lat: 23.5694S

Lon: 148.1756E

Elev: 624

StdP: 14.37

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	43.0	45.4	21.7	16.7	64.7	24.8	19.3	63.5	20.2	61.3	18.5	61.1	5.3	160	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.1	100.5	70.8	97.8	70.6	95.4	70.2	77.4	84.7	76.3	84.4	75.4	84.1	9.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.2	139.8	78.6	74.7	133.0	78.1	73.5	127.7	77.9	41.4	83.8	40.3	84.6	39.4	83.8	95.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	17.8	16.2	DB	37.8	105.2	2.2	3.1	36.2	107.4	34.9	109.2	33.6	111.0	32.0	113.3	(4)
(5)			WB	33.8	80.8	2.4	4.6	32.1	84.1	30.7	86.7	29.4	89.3	27.7	92.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	73.2	82.6	81.6	79.4	74.0	67.2	61.9	61.0	63.8	70.1	75.7	79.3	82.0	(6)
(7)		DBStd	8.95	3.82	4.15	3.62	3.88	4.53	4.77	4.39	4.62	4.67	4.65	4.27	4.53	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	374	0	0	0	2	29	115	139	78	10	1	0	0	(9)
(10)		CDD50	8457	1012	884	912	719	533	357	340	427	602	798	880	993	(10)
(11)		CDD65	3354	547	464	447	271	97	22	13	40	162	333	430	528	(11)
(12)		CDH74	35787	6338	5066	4564	2272	728	158	135	519	1798	3465	4632	6112	(12)
(13)		CDH80	16624	3265	2493	2076	766	118	8	6	100	635	1576	2318	3263	(13)
(14)	Wind	WSAvg	8.3	9.2	8.8	8.5	7.8	7.3	7.6	7.4	7.6	7.9	8.9	9.2	9.4	(14)
(15)	Precipitation	PrecAvg	25.4	3.8	3.8	2.4	1.4	2.0	0.8	1.1	0.8	0.7	1.5	2.6	4.4	(15)
(16)		PrecMax	42.3	16.7	10.5	12.2	6.6	10.2	4.4	6.2	5.2	3.3	4.9	5.7	11.3	(16)
(17)		PrecMin	13.5	0.2	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.9	(17)
(18)		PrecStd	6.9	3.1	2.7	2.4	1.6	2.4	1.0	1.4	1.3	0.9	1.5	1.6	2.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.6	102.9	98.7	92.2	86.7	80.7	80.6	87.0	94.9	100.1	101.2	104.1	(19)
(20)			MCWB	72.8	71.9	70.7	69.4	66.1	63.2	62.7	61.5	63.6	66.3	69.7	70.0	(20)
(21)		2%	DB	99.0	97.8	94.5	89.1	82.9	78.0	77.4	82.4	90.1	95.0	97.2	100.0	(21)
(22)			MCWB	72.5	72.6	70.9	67.6	64.0	62.2	60.2	61.0	63.0	66.2	68.6	70.4	(22)
(23)		5%	DB	95.9	94.7	92.1	86.8	80.6	75.7	75.0	79.3	86.6	91.5	94.4	96.8	(23)
(24)			MCWB	72.0	72.2	70.3	66.7	63.0	61.0	59.1	60.0	62.5	65.8	68.3	70.3	(24)
(25)		10%	DB	93.3	92.0	89.7	84.6	78.2	73.2	72.6	76.4	83.5	88.4	91.5	93.7	(25)
(26)			MCWB	71.8	71.7	69.6	66.0	62.3	59.7	58.1	58.6	61.6	65.2	67.6	69.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.6	78.9	77.5	74.7	71.0	68.9	67.5	67.6	71.9	73.5	75.4	77.4	(27)
(28)			MCD B	86.3	83.2	83.7	81.8	78.4	73.3	74.3	77.4	77.9	83.4	87.4	84.8	(28)
(29)		2%	WB	77.4	77.2	76.1	72.3	68.2	65.8	65.1	65.2	68.8	71.5	73.5	75.8	(29)
(30)			MCD B	85.9	84.1	82.6	80.5	76.9	72.4	71.7	75.0	77.0	82.1	84.9	85.3	(30)
(31)		5%	WB	76.2	76.1	74.8	70.8	66.4	63.9	62.6	63.2	66.6	70.0	72.1	74.7	(31)
(32)			MCD B	85.0	84.1	82.1	79.7	75.7	71.5	69.8	73.5	77.3	80.7	84.1	84.9	(32)
(33)		10%	WB	75.2	75.2	73.6	69.3	64.8	62.1	60.5	61.3	64.8	68.7	71.0	73.6	(33)
(34)			MCD B	83.9	83.9	82.1	78.4	74.1	69.9	68.5	72.2	77.3	80.3	83.0	83.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.1	19.3	20.3	21.3	22.1	21.8	23.9	25.9	26.3	24.4	22.6	21.4	(35)
(36)			MCDBR	24.4	23.4	23.4	23.5	23.8	24.0	25.9	29.2	30.6	28.9	26.7	25.8	(36)
(37)			MCWBR	6.1	6.0	6.0	8.0	9.9	11.1	12.4	12.3	10.6	8.8	7.5	6.8	(37)
(38)		5% WB	MCD BR	18.8	17.7	17.3	18.7	19.5	19.1	20.5	24.4	22.8	22.5	21.7	20.4	(38)
(39)			MCWBR	6.4	6.1	6.1	7.2	8.4	9.4	10.4	10.6	9.3	8.2	7.3	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.386	0.372	0.352	0.336	0.316	0.306	0.297	0.311	0.345	0.373	0.389	0.390	(40)	
(41)		taud	2.508	2.552	2.597	2.605	2.600	2.638	2.637	2.576	2.482	2.438	2.435	2.468	(41)	
(42)		Ebn at Noon	304	304	302	293	287	284	291	297	299	300	301	303	(42)	
(43)		Edh at Noon	36	34	31	29	26	24	25	29	34	38	39	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2165	2063	1872	1647	1378	1184	1331	1628	1938	2180	2253	2262	(44)	
(45)		RadStd	154	179	152	148	71	93	97	98	171	136	246	205	45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page